
Shunting Inhibition and Dendritic Branching Shape Local Credit Assignment

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Biological neurons assign credit through structures that standard neural networks
2 usually ignore: synapses are distributed across branching dendrites, excitatory
3 and inhibitory inputs evoke synaptic conductances, and inhibition can shunt local
4 voltage by changing total input resistance. We study conductance-based dendritic
5 networks with E/I synapse banks, shunting inhibition, and tree-structured branch-
6 to-soma coupling, and ask when a low-bandwidth somatic broadcast can replace
7 compartment-specific backpropagated errors. Exact gradients for these dendritic
8 trees show that each synaptic update separates into a local eligibility term built
9 from presynaptic activity, driving force, and input resistance, and one non-local
10 compartment error. Under identity upward transfer, that error is the somatic error
11 transported through conductance-stage path gains. This factorization turns local
12 learning into a credit-signal compression problem: broadcast is useful only when it
13 approximates the path-specific compartment-error field. We test this mechanism
14 across gradient, oracle-transport, inhibition-intervention, competence, morphology,
15 and harder-data diagnostics. Shunting improves LocalCA–backprop alignment,
16 reduces gradient-scale mismatch, and makes exact compartment errors more rank-1
17 compressible than matched additive controls in the regimes studied here. Inhibitory-
18 conductance interventions show that learned inhibition carries sample-specific
19 pathway structure, while transported-error oracle experiments show that restoring
20 the exact path-gain error largely closes the local-learning gap. Under nonnegative
21 conductances and rank-1 broadcast, shunting LocalCA remains within about 5–6
22 percentage points of matched backpropagation reference performance on MNIST,
23 Fashion-MNIST, and context gating. These results suggest that E/I conductance,
24 shunting inhibition, and dendritic branching can reshape credit-signal geometry
25 and make low-bandwidth local learning more faithful.

1 Introduction

26 Unlike point units in standard neural networks, biological neurons assign credit across branching
27 dendritic trees. A synapse does not only see presynaptic activity and a scalar postsynaptic activation;
28 it also has access to local membrane voltage, synaptic reversal potential, total conductance, and
29 input resistance. Inhibitory input can therefore do more than subtract activity: by increasing local
30 conductance, it can shunt voltage and change the gain of every dendritic path passing through that
31 compartment.

32 We examine whether these biophysical ingredients can facilitate local credit assignment (LocalCA).
33 Backpropagation would assign a distinct error to every dendritic compartment, whereas a biologically
34 plausible supervisory signal is more likely to arrive as a low-bandwidth somatic or modulatory
35 broadcast [14]. The central question is therefore not whether such a broadcast can reproduce
36

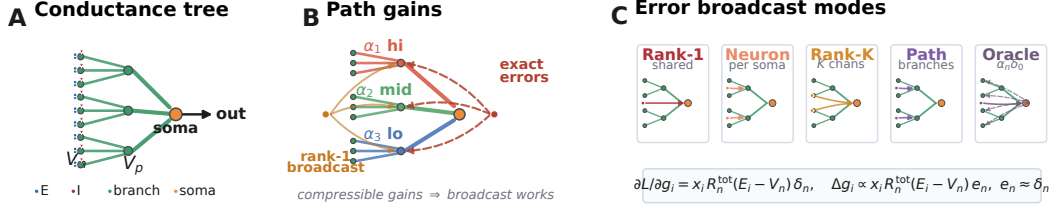


Figure 1: Model and credit assignment. (A) Conductance tree with nonnegative excitatory drive and inhibitory shunting load. (B) Exact credit varies by dendritic path, whereas rank-1 LocalCA broadcasts one shared error field. (C) LocalCA preserves the exact synapse-local eligibility and substitutes $e_n \approx \delta_n$; rank-1, neuron-wise, rank- K , and path-structured broadcasts trade biological bandwidth for fidelity, while transported error is an oracle.

unconstrained backpropagation in every setting, but when the exact dendritic error field is simple enough for it to approximate. Our contribution is not simply to add dendritic structure to a neural network, but to show how conductance, shunting inhibition, and dendritic topology change the geometry of the credit signal itself.

Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for dendritic trees (Theorem 1). Each synaptic gradient factorizes into a synapse-local eligibility term and a single non-local compartment error. The eligibility term depends on presynaptic activity, driving force, and input resistance. The compartment error is path-specific: under identity upward transfer it is the somatic error multiplied by a conductance-stage path gain through the dendritic tree. LocalCA preserves the exact local eligibility and replaces this path-specific error with a broadcast field.

This factorization turns local credit assignment into a compressibility problem. By compressibility, we mean that the matrix of exact compartment errors across examples and compartments is well approximated by a low-rank signal, especially a rank-1 field shared across compartments. Dendritic branching creates the paths; E/I synapse banks set the local conductance state; and shunting inhibition changes the input resistance along each path. Shunting is not assumed to help monotonically. It helps rank-1 broadcast when inhibitory conductance attenuates high-gain paths or otherwise makes the exact error field more compressible.

Related work. This work builds on models of dendritic integration, branch-level nonlinear computation, normalization, and biologically plausible teaching signals [1, 3–6, 11–13, 16, 22–24, 30, 31, 44]. It also connects to machine-learning approaches that replace exact backpropagation with local, random, predictive, equilibrium, perturbation, or broadcast feedback signals [9, 10, 15, 25–29, 32–36]. Our contribution is complementary: we derive the exact conductance-tree credit object and test when shunting inhibition and dendritic morphology make that object broadcast-compatible.

We test this chain directly. Exact-gradient reconstruction verifies the factorization numerically. Gradient-fidelity diagnostics show that shunting produces local updates that are better aligned and better scaled relative to backpropagation than matched additive controls. Exact-error compressibility and inhibition-intervention diagnostics show that learned inhibitory conductance carries sample-specific pathway structure. Transported-error oracle experiments show that supplying the exact path-gain-transported compartment error largely closes the local-learning gap, identifying broadcast fidelity as the main bottleneck. In an MNIST gradient-dynamics diagnostic, shunting improves final LocalCA–backprop alignment (cosine 0.100 ± 0.033 vs. 0.005 ± 0.026) and reduces scale mismatch (0.26 ± 0.05 vs. 0.50 ± 0.05). On noise resilience under matched inhibition ($N_I \geq 5$), transported error raises additive local learning from 81–84% under rank-1 broadcast to 92–94%, nearly matching shunting at 95–95.2%.

Contributions. This paper makes three contributions: deriving the exact conductance-tree gradient and its local-eligibility/path-error factorization; identifying path-gain compressibility as the quantity mediating whether low-rank somatic broadcast can approximate that error; and showing empirically that shunting inhibition can improve this approximation while also identifying regimes where rank-1 feedback fails and richer pathway-specific feedback is needed.

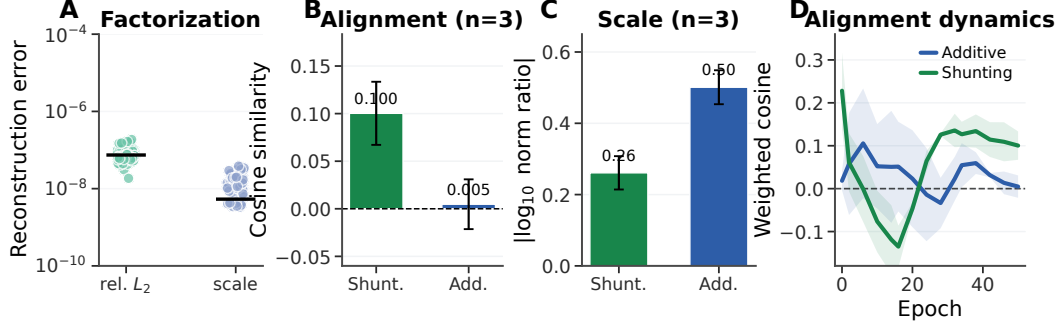


Figure 2: **Gradient-fidelity diagnostic.** (A) Exact-factorization sanity check: reconstructed gradients match autograd to numerical precision. (B) Final-state weighted cosine similarity between local and backprop gradients. (C) Final-state scale mismatch, $|\log_{10}(\|g^{\text{local}}\|/\|g^{\text{bp}}\|)|$. Error bars are ± 1 s.d. across 3 gradient-dynamics seeds; main learning and oracle sweeps use 5 seeds unless noted. (D) Alignment over learning; Appendix Fig. S2 shows finite gradient norms.

2 Compartmental Voltage Model and Gradient Derivation

We use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal potentials are set to 0, the excitatory reversal potential is set to 1, and leak conductance is fixed at 1 (i.e., all conductances are expressed relative to leak). Each compartment computes a conductance-weighted voltage: excitatory conductance pulls the voltage toward the excitatory reversal potential, leak and shunting inhibition pull it toward zero, and dendritic coupling transfers child voltages upward toward the soma. This form makes local sensitivity depend on both driving force ($E - V$) and total input resistance R^{tot} . Throughout the paper we write dendritic morphologies as rooted trees $[b_1, b_2, \dots, b_D]$, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. Thus $[3, 3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma. In the main feedforward experiments, synaptic inputs terminate on dendritic branches rather than directly on the soma. Each branch layer carries explicit excitatory conductance banks and, when enabled, learned inhibitory branch-level conductance banks driven by nonnegative inputs. This is input-driven shunting inhibition, not a recurrent lateral inhibitory circuit; explicit inhibitory-cell controls are reported in the appendix. The additive and shunting cores are architecture-matched at the tree level and differ only in the branch-level voltage integration rule. These ingredients play distinct roles in the credit-assignment argument. Branching creates the path structure. E/I synapse banks determine the local conductance state. Shunting turns inhibitory conductance into a multiplicative change in path gain through input resistance. The local learning rule then asks whether a low-bandwidth teaching signal can stand in for the exact path-specific error induced by that tree. **Notation.** We use α_n for the exact conductance-stage path gain from compartment n to the soma, r_n^{4F} for the empirical 4F branch-soma covariance proxy, ϕ_n for the 5F confidence factor, λ_{mix} for the pathway-broadcast mixing weight, and κ_j for depth-dependent modulation in morphology extensions. All conductances and presynaptic activities in the main experimental setting are nonnegative; additive voltages may be signed comparators, but the conductance variables that define the shunting model are nonnegative.

Voltage equation and local sensitivities. Consider compartment n with synaptic inputs j (activity x_j , reversal E_j , conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j , conductance $g_j^{\text{den}} \geq 0$). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \quad (1)$$

V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathcal{S}_n = \{0\} \cup \{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$ and $0 < R_n^{\text{tot}} \leq 1$. The local sensitivities follow directly:

Proposition 1 (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}}(E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}}(V_i - V_n). \quad (2)$$

Eq. (2) gives the eligibility factors used by the local rules below: presynaptic activity or voltage difference, driving force, and input resistance.

Shunting inhibition as divisive gain control. An inhibitory synapse with $E_{\text{inh}} \approx 0$ contributes current $(0 - V_n)x_j g_j^{\text{syn}}$ and increases g_n^{tot} . Its sensitivity is $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$, which corresponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive in some regimes [7]. Inhibitory plasticity can balance excitation dynamically [8]; our learned inhibitory conductances serve an analogous role. The same denominator also gives the direct connection to path gain. Writing $G_n^E(x)$ and $G_n^I(x)$ for the total excitatory and inhibitory synaptic conductance impinging on compartment n ,

$$R_n^{\text{tot}}(x) = (1 + G_n^E(x) + G_n^I(x) + G_n^{\text{den}})^{-1}.$$

Increasing $G_n^I(x)$ lowers $R_n^{\text{tot}}(x)$ and therefore multiplicatively suppresses every upstream path whose error must pass through compartment n . In this sense inhibition can gate credit flow by changing the path gain, even when the inhibitory drive is feedforward rather than lateral. Prop. 2 formalizes this statement after the tree path gain is defined.

Exact gradients for dendritic trees. Let V_0 be the somatic output and define the soma/core error as $\delta_0 := \partial L / \partial V_0$. For a linear decoder $\hat{y} = W_{\text{dec}} V_0$, this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$; for a nonlinear decoder, δ_0 is the corresponding decoder-input Jacobian product. The theoretical derivation is stated for the conductance-stage voltage before any post-voltage reactivation. When a branch applies a monotone reactivation before transmitting activity upward, the same recursion holds with additional local derivative factors along the path. All exact-error diagnostics below are computed on pre-activation voltages, so they test the conductance-stage quantity derived here.

Theorem 1 (Conductance-Stage Backpropagation on a Dendritic Tree). *For a rooted dendritic tree with soma at node 0 and unique parent $p(n)$ for each non-somatic compartment, suppose the parent receives either the pre-activation voltage V_n or a local post-voltage activity $a_n = f_n(V_n)$. For identity upward transfer, the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

with boundary condition $\partial L / \partial V_0 = \delta_0$. With post-voltage transfer $a_n = f_n(V_n)$, the local factor becomes $f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}$. Because the graph is a tree, each compartment has a unique path to the soma. Defining the conductance-stage path gain

$$\alpha_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n \delta_0, \quad (4)$$

with $\alpha_0 = 1$ under identity transfer. In the post-voltage case, the exact effective path gain is obtained by multiplying the same product by the local factors $f'_i(V_i)$ on the transmitted child activities along the path.

Proof. Apply the chain rule on the tree-structured computation graph using Prop. 1. $\square$

The one-step recursion in Eq. (3) and the path product in Eq. (4) separate local conductance transfer from the non-local somatic error.

Proposition 2 (Inhibitory control of conductance-stage path gain). *Let $\mathcal{A}(n)$ be the set of compartments on the path from compartment n to the soma, excluding n and including the parent compartments through which the error must pass. Holding dendritic coupling conductances fixed, an added inhibitory conductance $\Delta G_k^I(x) \geq 0$ at any $k \in \mathcal{A}(n)$ changes the conductance-stage path gain by*

$$\frac{\alpha_n(x; \Delta G^I)}{\alpha_n(x; 0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}(x)}{g_k^{\text{tot}}(x) + \Delta G_k^I(x)}, \quad \frac{\partial \log \alpha_n}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n). \end{cases} \quad (5)$$

146 *Proof.* Substitute $R_k^{\text{tot}} = 1/g_k^{\text{tot}}$ into the path-gain product in Eq. (4). Adding inhibitory conductance
 147 changes only the denominator at compartments on the path from n to the soma, which gives the
 148 product ratio and the log derivative in Eq. (5). $\square$

149 Prop. 2 makes two directed predictions. First, inhibition is a path gate: it attenuates all credit paths
 150 below the inhibited compartment. Second, inhibition is useful for rank-1 local learning only when this
 151 attenuation makes the distribution of α_n more uniform or better aligned with the task. If inhibition
 152 is too strong, too heterogeneous, or generated by a poorly calibrated inhibitory population, it can
 153 suppress useful signals and hurt learning. In the additive control the same I input changes voltages
 154 and local eligibilities, but it does not enter the conductance-stage path recursion as a multiplicative
 155 resistance gate.

156 **When does shunting compress path gains?** The attenuation identity above is not by itself a
 157 concentration result. Let $z_n = \log \alpha_n$ be the log path gain for compartment n . If added inhibitory
 158 conductance contributes a path-dependent attenuation

$$\beta_n(x) = \sum_{k \in \mathcal{A}(n)} \log \left(1 + \frac{\Delta G_k^I(x)}{g_k^{\text{tot}}(x)} \right), \quad z'_n = z_n - \beta_n, \quad (6)$$

159 then, over compartments or examples,

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2 \text{Cov}(z, \beta). \quad (7)$$

160 Thus shunting narrows the log path-gain field exactly when

$$\text{Cov}(z, \beta) > \frac{1}{2} \text{Var}(\beta). \quad (8)$$

161 Eq. (6) follows by taking the logarithm of the product ratio in Prop. 2; Eq. (7) then applies the variance
 162 identity for $z - \beta$. Eq. (8) is an interpretive sufficient condition under the same averaging measure,
 163 not a primary empirical diagnostic. In finite trained networks, the covariance margin can disagree
 164 with CV and exact-error-rank summaries because these statistics average over different compartment
 165 and sample axes. We therefore use direct path-gain dispersion, exact-error compressibility, and
 166 inhibition-intervention measurements as the empirical tests, and report the covariance margin as a
 167 boundary check in the appendix.

168 **Corollary 1** (Local–Global Factorization). *The exact synaptic gradient at compartment n factorizes*
 169 *as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (9)$$

170 *where the eligibility term depends only on quantities available at synapse i (presynaptic activity*
 171 *x_i , input resistance R_n^{tot} , driving force $E_i - V_n$), and the compartment error $\partial L / \partial V_n$ is the sole*
 172 *non-local quantity.*

173 The factorization in Eq. (9) isolates the only non-local quantity as $\partial L / \partial V_n = \alpha_n \delta_0$ under identity
 174 upward transfer. With post-voltage transfer, the activation-derivative factors in Theorem 1 enter the
 175 effective path gain, but the compartment error remains the only non-local term. Any practical local
 176 rule therefore depends entirely on how well a broadcast signal approximates that exact compartment
 177 error, which we evaluate in Sec. 4. In the models studied here, shunting improves this approximation
 178 by reducing and stabilizing R_n^{tot} , which narrows the spread of local sensitivities across the tree. In
 179 the empirical networks, each dendritic branch layer may optionally apply a post-voltage reactivation
 180 $f(V_n)$ before passing activity to the next stage; unless otherwise noted, the manuscript sweeps use a
 181 learnable bounded `param_tanh` reactivation.

182 3 Local Learning Rules

183 **Broadcast error approximation.** We approximate the exact compartment error $\partial L / \partial V_n$ (Corol-
 184 lary 1) with a broadcast signal e_n derived from the somatic/core error δ_0 . For a linear readout
 185 this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$; for a nonlinear decoder we use the corresponding decoder-
 186 input Jacobian product $J_{\text{dec}}(h_{\text{core}})^\top (\partial L / \partial \hat{y})$ so that the broadcast remains defined in the same

core-space coordinates. We distinguish five feedback objects: **rank-1 shared broadcast**, $e_n = \bar{\delta}\mathbf{1}$ with $\bar{\delta} = \text{mean}(\delta_0)$; **neuron-wise broadcast**, $e_n = \delta_0$ when layer width matches the error dimension; **rank- K random broadcast**, $c = P_K\delta_0$ and $e_n = \sum_k q_{n,k}c_k$; **path-structured broadcast**, $e_n = \lambda_{\text{mix}}\bar{\delta}\mathbf{1} + (1 - \lambda_{\text{mix}})\Gamma_n(\delta_0)$ using router-inferred pathway roles [43]; and the **transported oracle**, $e_n^{\text{transport}} = \alpha_n\delta_0 = \partial L/\partial V_n$, an analysis upper bound because it supplies the exact path gain from Theorem 1. In the main feedforward architectures, hidden widths exceed the decoder-error dimension, so the default neuron-wise broadcast reduces to a rank-1 shared field (Appendix Table S11). Thus the main rank-1 result should be read as the architecture-induced low-bandwidth case, not as a claim that neuron-wise broadcasts were unavailable in principle. Under shunting, that deliberately low-bandwidth field is often sufficient on standard classification tasks. Routed tasks and rank-bridge controls in the appendix test the opposite regime, where rank-1 broadcast collapses branch identity and richer feedback is needed. A local-mismatch heuristic is included only as an appendix control because it performs much worse than the somatic-error broadcasts (Appendix A).

200 Three-factor learning rule (3F).

201 **Definition 1** (3F Update). *For synaptic and dendritic conductances:*

$$\Delta g_j^{\text{syn}} = \eta \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \Delta g_j^{\text{den}} = \eta \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B, \quad (10)$$

202 where $\langle \cdot \rangle_B$ denotes the batch average.

203 The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error e_n . The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force ($E_j - V_n$). We write Δg as the local gradient estimate supplied to the optimizer; the optimizer applies the usual descent step. Equivalently, one could absorb the minus sign into e_n and treat it as a descent signal.

204 **Additive control.** Rule (10) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For the additive control ($V_n = \sum g_j x_j$, no divisive normalization), the correct local gradient is simpler: $\Delta g_j^{\text{syn}} = \eta \langle x_j e_n \rangle_B$, with no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model.

214 **Practical heuristic wrappers: 4F and 5F. 4F (branch-soma relevance proxy).** In the conductance-stage recursion (4), the path gain α_n attenuates the somatic error differently at each compartment. To compensate without explicitly computing that gain, 4F uses an online correlation proxy

$$r_n^{4F} = \text{Cov}(\bar{V}_n, \bar{V}_0) / (\sqrt{\text{Var}(\bar{V}_n) \text{Var}(\bar{V}_0)} + \varepsilon).$$

217 High r_n^{4F} up-weights branches whose voltages covary with the soma; low r_n^{4F} down-weights branches where the broadcast is likely to be a poor proxy. We keep 4F as a diagnostic intermediate. Empirically, this factor alone gives little improvement over 3F; the practical jump comes from the 5F confidence factor below.

221 **5F (confidence-weighted 4F).** 5F adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \varepsilon),$$

222 where σ_{res}^2 is the residual variance of V_n after linear regression on the parent voltage. Operationally, ϕ_n is a branch-wise confidence gate, clamped to $[0.25, 4.0]$ for stability. This factor is empirical, not theorem-derived; its gain should be read as practical denoising rather than theoretical necessity.

Definition 2 (5F Update).

$$\Delta g_j^{\text{syn}} = \eta r_n^{4F} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B. \quad (11)$$

225 Eq. (11) is therefore the main practical LocalCA update, while the 3F rule is the theoretically derived local gradient and 4F is the intermediate ablation. A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive but not central to the main mechanism claim.

228 4 Experiments

229 **Setup.** The experiments test five linked predictions: exact factorization should reconstruct autograd gradients; shunting should improve LocalCA direction and scale relative to backpropagation; exact

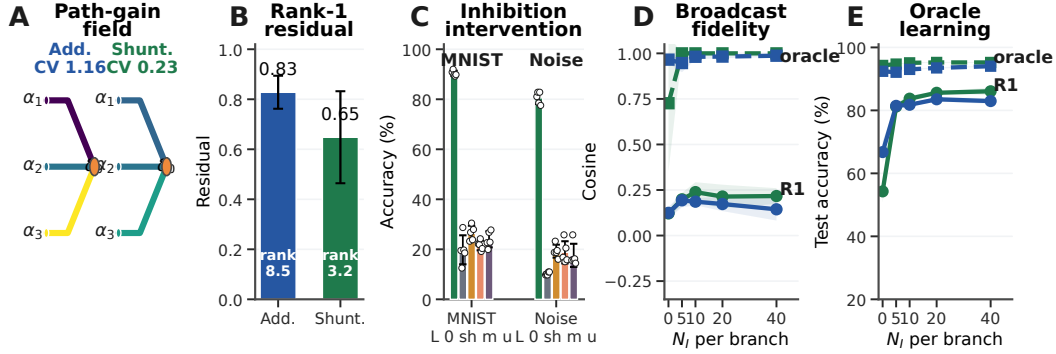


Figure 3: **Mechanistic chain from path gains to learning.** (A) Dendritic path-gain fields at matched inhibition. (B) Exact-error compressibility on MNIST: lower rank-1 residual and participation rank favor scalar broadcast. (C) Post-training inhibitory-conductance interventions (L: learned; 0/sh/m/u: zero, shuffled, mean-clamped, uniform matched). (D) Compartment-error fidelity on noise resilience; dashed lines show the transported-error oracle. (E) Transported-error learning recovers near-ceiling local learning where rank-1 broadcast fails. Error bars denote ± 1 s.d. across seeds.

compartment errors should be more compressible under shunting than under matched additive controls; learned inhibition should carry sample-specific pathway structure rather than merely adding conductance load; and replacing the broadcast with transported exact error should largely close the local-learning gap. We evaluate capacity-calibrated supervised tasks and controlled sweeps that isolate inhibition strength, rule family, broadcast mode, morphology, and architecture. The main text uses MNIST, Fashion-MNIST [37], context gating, noise resilience, and a compressed morphology-regime summary; CIFAR-10 and routed-feedback diagnostics are reported in the appendix (Figs. S6, S10). Main comparisons are architecture-matched additive vs. shunting dendritic cores and local vs. backprop training; DFA is a supplementary baseline. Unless otherwise stated, local runs use 5F with the default neuron-wise broadcast, which reduces to rank-1 shared broadcast when hidden width exceeds the decoder-error dimension. All main-claim dendritic runs use nonnegative conductance transforms and nonnegative first-layer synaptic drive; image inputs are in $[0, 1]$, corrupted inputs are clipped back to $[0, 1]$, and context gating uses a nonnegative contextual figure-ground MNIST construction. The transported-error condition is an oracle upper bound derived from Theorem 1, not the proposed biological rule. Main mechanistic, oracle, and competence results use 5 seeds per condition unless noted otherwise.

From exact factorization to credit-signal fidelity.

To test whether these performance differences correspond to better credit signals, we compare local and backprop gradients on the same batch at the same parameter values. For each parameter tensor p , we compute directional alignment (cosine similarity) and scale mismatch, defined as $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$, aggregated by parameter count.

Across the MNIST gradient-dynamics diagnostic, shunting is more aligned with backprop and closer in norm than the additive control at the final checkpoint: weighted cosine rises from 0.005 ± 0.026 to 0.100 ± 0.033 , and scale mismatch falls from 0.50 ± 0.05 to 0.26 ± 0.05 . Figure 2A confirms that the implementation matches the theory under the nonnegative-input setup: reconstructing gradients from Corollary 1 using exact compartment errors yields weighted relative reconstruction error below 2×10^{-7} and weighted scale mismatch below 4×10^{-8} across runs. In a few low-norm parameter blocks, cosine becomes numerically unstable even though the reconstruction itself is essentially exact, so we report the norm-based diagnostics in the figure. Figure 2D shows the same gradient-alignment comparison over training: additive alignment remains near zero at the final checkpoint rather than failing only at one static snapshot. Appendix Fig. S2 verifies that the corresponding local and backprop gradient norms remain finite throughout learning, ruling out a trivial zero-gradient explanation. At the level of the theoretically derived 3F rule, the central non-local approximation is therefore the broadcast substitute for $\partial L/\partial V_n$.

Mechanistic chain: path gains, broadcast fidelity, and oracle transport. Figure 3 provides the central empirical chain. At matched inhibition ($N_I=5$), shunting narrows the MNIST conductance-

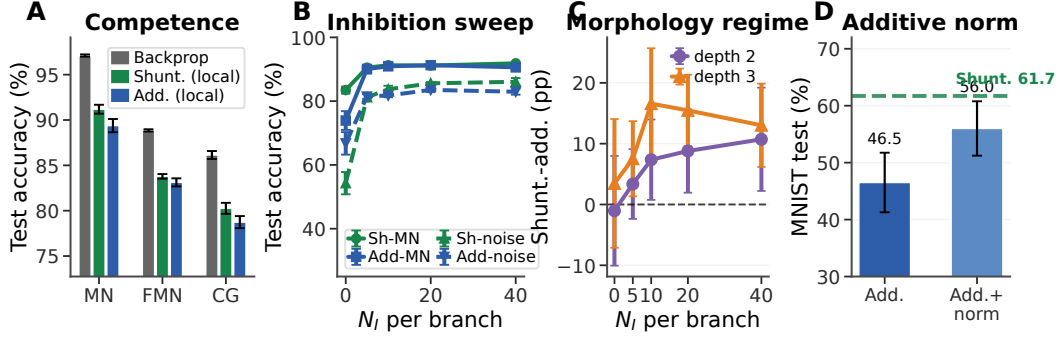


Figure 4: **Capacity-calibrated competence and regime dependence.** (A) Matched backprop reference performance vs. rank-1 5F LocalCA on MNIST, Fashion-MNIST, and context gating. (B) Inhibitory dose-response. (C) Morphology-dependent shunting advantage on noise resilience from a 3-seed sweep. (D) Additive normalization improves additive LocalCA but remains below the shunting reference. Error bars denote ± 1 s.d. where available.

stage path-gain field (CV 0.23 vs. 1.16 for additive) and makes the exact compartment-error matrix more compressible: the best rank-1 residual drops from 0.83 ± 0.07 to 0.65 ± 0.18 , while participation rank drops from 8.5 ± 3.3 to 3.2 ± 1.2 . This is the diagnostic matched to the paper’s main claim: rank-1 broadcast is useful when the exact error field is closer to low rank.

The inhibition intervention shows that the learned inhibitory field is necessary for the trained computation and is not interchangeable with a matched conductance load: zeroing, shuffling, or clamping learned branch inhibition drops MNIST from about 90% to 20–26% and noise resilience from about 81% to 10–19%. The fidelity and oracle panels test the same bottleneck: rank-1 broadcast tracks the exact compartment error better under shunting once inhibition is present, transported error reaches about 0.95–1.0 fidelity, and transported learning lifts both cores to 92–95% on noise resilience. CIFAR-10 and direct-I versus explicit-I path-gain probes remain appendix diagnostics (Fig. S6; Table S4).

Capacity-calibrated competence. In the capacity-calibrated regime, shunting LocalCA remains within about 5–6 percentage points of matched backpropagation on MNIST, Fashion-MNIST, and context gating (Table S2; Fig. 4). Figure 4D shows that additive normalization improves additive LocalCA but remains below the shunting reference, so the gain is not explained by generic voltage normalization alone. Figure 5 summarizes the rule and feedback controls: 5F is the strongest practical optimizer, somatic rank-1 error is essential, and higher feedback bandwidth improves the noise-resilience diagnostic.

Regime dependence of the shunting advantage. With matched rank-1 broadcast, shunting is modestly better on MNIST, Fashion-MNIST, and context gating, including 0.803 ± 0.006 vs. 0.787 ± 0.007 for additive on context gating. The gap grows on inhibition-sensitive noise resilience once inhibitory conductance is present (Fig. 4B). Morphology shifts the useful inhibitory operating regime: different tree depths and branch factors peak at different N_I , and the shunting advantage is not monotonic in inhibition (Fig. 4C). Appendix Fig. S4 extends this boundary check to depth scaling and noisy error broadcasts.

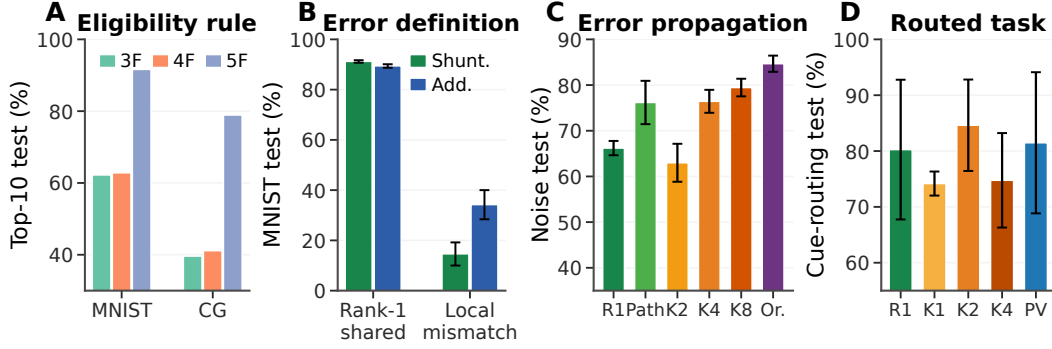


Figure 5: **Rule and feedback controls.** (A) The practical gain comes mainly from the 5F confidence wrapper; 3F/4F remain weak optimizers. (B) Replacing the somatic rank-1 error with a local-mismatch heuristic collapses learning. (C) Richer or transported feedback improves noise resilience. (D) Cue routing probes a branch-specific regime: higher rank helps, while the current pathway-vector implementation does not yet beat the best unstructured low-rank control.

5 Discussion and Limitations

We studied local credit assignment in dendritic networks with E/I conductance synapses, shunting inhibition, and tree-structured branching. Exact gradients separate synapse-local eligibility-presynaptic activity, driving force, and input resistance—from one path-specific compartment error. This makes LocalCA an error-approximation problem: the local factors are available at the synapse, but the teaching signal must approximate the exact dendritic error field.

The experiments support a coherent mechanism. Branching creates credit paths, E/I banks set each compartment’s conductance state, and shunting modulates input resistance along those paths. In the regimes studied here, this makes exact errors more rank-1 compressible, improves LocalCA-backprop alignment, and reduces scale mismatch. Transported-error oracles largely close the local-learning gap, while inhibition interventions show that learned inhibitory conductance carries sample-specific pathway structure rather than only global conductance scale.

The limits are important. Shunting is not universal, and more inhibition is not always better. The practical 5F rule is an empirical stabilizer rather than a theorem-derived necessity; Eq. (8) is interpretive; activation shape, morphology, task structure, and broadcast rank all change the operating regime. The main model also uses input-driven shunting inhibition rather than recurrent lateral inhibition, so recurrent inhibitory circuits remain a future direction. The broader message is that biophysical dendrites can change not only representations, but also the geometry of credit assignment.

References

- [1] Koch, C. (1999). *Biophysics of Computation*. Oxford University Press.
- [2] Dayan, P., & Abbott, L. F. (2001). *Theoretical Neuroscience*. MIT Press.
- [3] Poirazi, P., Brannon, T., & Mel, B. W. (2003). Pyramidal neuron as two-layer neural network. *Neuron*, 37(6), 989–999.
- [4] London, M., & Häusser, M. (2005). Dendritic computation. *Annual Review of Neuroscience*, 28, 503–532.
- [5] Urbanczik, R., & Senn, W. (2014). Learning by the dendritic prediction of somatic spiking. *Neuron*, 81(3), 521–528.
- [6] Carandini, M., & Heeger, D. J. (2012). Normalization as a canonical neural computation. *Nature Reviews Neuroscience*, 13(1), 51–62.
- [7] Holt, G. R., & Koch, C. (1997). Shunting inhibition does not have a divisive effect on firing rates. *Neural Computation*, 9(5), 1001–1013.

- [8] Vogels, T. P., Sprekeler, H., Zenke, F., Clopath, C., & Gerstner, W. (2011). Inhibitory plasticity balances excitation and inhibition in sensory pathways and memory networks. *Science*, 334(6062), 1569–1573.
- [9] Lillicrap, T. P., Cownden, D., Tweed, D. B., & Akerman, C. J. (2016). Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7, 13276.
- [10] Nøkland, A. (2016). Direct feedback alignment provides learning in deep neural networks. *NeurIPS*, 29.
- [11] Guerguiev, J., Lillicrap, T. P., & Richards, B. A. (2017). Towards deep learning with segregated dendrites. *eLife*, 6, e22901.
- [12] Sacramento, J., Costa, R. P., Bengio, Y., & Senn, W. (2018). Dendritic cortical microcircuits approximate the backpropagation algorithm. *NeurIPS*, 31.
- [13] Whittington, J. C., & Bogacz, R. (2019). Theories of error back-propagation in the brain. *Trends in Cognitive Sciences*, 23(3), 235–250.
- [14] Richards, B. A., & Lillicrap, T. P. (2019). Dendritic solutions to the credit assignment problem. *Current Opinion in Neurobiology*, 54, 28–36.
- [15] Scellier, B., & Bengio, Y. (2017). Equilibrium propagation. *Frontiers in Computational Neuroscience*, 11, 24.
- [16] Song, Y., Millidge, B., Salvatori, T., Lukasiewicz, T., Xu, Z., & Bogacz, R. (2024). Inferring neural activity before plasticity as a foundation for learning beyond backpropagation. *Nature Neuroscience*, 27, 348–358.
- [17] Gretton, A., Bousquet, O., Smola, A. J., & Schölkopf, B. (2005). Measuring statistical dependence with Hilbert-Schmidt norms. *Algorithmic Learning Theory*, 63–77.
- [18] Frémaux, N., & Gerstner, W. (2016). Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9, 85.
- [19] Bellec, G., et al. (2020). A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 11, 3625.
- [20] Welford, B. P. (1962). Note on a method for calculating corrected sums of squares and products. *Technometrics*, 4(3), 419–420.
- [21] Turrigiano, G. G. (2008). The self-tuning neuron: synaptic scaling of excitatory synapses. *Cell*, 135(3), 422–435.
- [22] Payeur, A., Guerguiev, J., Zenke, F., Richards, B. A., & Naud, R. (2021). Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24(7), 1010–1019.
- [23] Greedy, W., Zhu, H. W., Pemberton, J., Mellor, J., & Ponte Costa, R. (2022). Single-phase deep learning in cortico-cortical networks. *NeurIPS*, 35, 24213–24225.
- [24] Haider, P., Ellenberger, B., Kriener, L., Jordan, J., Senn, W., & Petrovici, M. A. (2021). Latent Equilibrium: A unified learning theory for arbitrarily fast computation with arbitrarily slow neurons. *NeurIPS*, 34.
- [25] Hinton, G. (2022). The forward-forward algorithm. *arXiv:2212.13345*.
- [26] Dellaferrera, G., & Kreiman, G. (2022). Error-driven input modulation: Solving the credit assignment problem without a backward pass. *Proceedings of the 39th International Conference on Machine Learning, Proceedings of Machine Learning Research*, 162, 4937–4955.
- [27] Lee, D.-H., et al. (2015). Difference target propagation. *ECML*, 498–515.

- [28] Meulemans, A., et al. (2021). Credit assignment in neural networks through deep feedback control. *NeurIPS*, 34.
- [29] Millidge, B., Seth, A. K., & Buckley, C. L. (2021). Predictive coding: A theoretical and experimental review. *arXiv:2107.12979*.
- [30] Koch, C., Poggio, T., & Torre, V. (1983). Nonlinear interactions in a dendritic tree. *PNAS*, 80(9), 2799–2802.
- [31] Silver, R. A. (2010). Neuronal arithmetic. *Nature Reviews Neuroscience*, 11(7), 474–489.
- [32] Max, K., Kriener, L., Pineda García, G. et al. (2024). Learning efficient backprojections across cortical hierarchies in real time. *Nature Machine Intelligence*, 6(6), 619–630.
- [33] Ma, C., Wu, J., Si, C., & Tan, K. C. (2024). Scaling supervised local learning with augmented auxiliary networks. *International Conference on Learning Representations (ICLR)*.
- [34] Lv, C., Xu, J., Lu, Y., Wang, X., Wang, Z., Xu, Z., Yu, D., Du, X., Zheng, X., & Huang, X. (2025). Dendritic Localized Learning: Toward Biologically Plausible Algorithm. *Proceedings of the 42nd International Conference on Machine Learning, Proceedings of Machine Learning Research*, 267, 41682–41700.
- [35] Erdogan, M., Pehlevan, C., & Erdogan, A. T. (2025). Error broadcast and decorrelation as a potential artificial and natural learning mechanism. *OpenReview*, NeurIPS 2025 spotlight.
- [36] Kao, C.-H., & Hariharan, B. (2024). Counter-Current Learning: A Biologically Plausible Dual Network Approach for Deep Learning. *Advances in Neural Information Processing Systems*, 37.
- [37] Xiao, H., Rasul, K., & Vollgraf, R. (2017). Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. *arXiv:1708.07747*.
- [38] Song, S., Sjöström, P. J., Reigl, M., Nelson, S., & Chklovskii, D. B. (2005). Highly nonrandom features of synaptic connectivity in local cortical circuits. *PLoS Biology*, 3(3), e68.
- [39] Buzsáki, G., & Mizuseki, K. (2014). The log-dynamic brain: How skewed distributions affect network operations. *Nature Reviews Neuroscience*, 15(4), 264–278.
- [40] The MICrONS Consortium (2025). Functional connectomics spanning multiple areas of mouse visual cortex. *Nature*, 640(8058), 435–447.
- [41] Loewenstein, Y., Kuras, A., & Rumpel, S. (2011). Multiplicative Dynamics Underlie the Emergence of the Log-Normal Distribution of Spine Sizes in the Neocortex In Vivo. *Journal of Neuroscience*, 31(26), 9481–9488.
- [42] Megías, M., et al. (2001). Total number and distribution of inhibitory and excitatory synapses on hippocampal CA1 pyramidal cells. *Neuroscience*, 102(3), 527–540.
- [43] Francioni, V., Tang, V. D., Toloza, E. H. S. et al. (2026). Vectorized instructive signals in cortical dendrites. *Nature*. <https://doi.org/10.1038/s41586-026-10190-7>
- [44] Iyer, A., Grewal, K., Velu, A., Souza, L. O., Forest, J., & Ahmad, S. (2022). Avoiding catastrophe: Active dendrites enable multi-task learning in dynamic environments. *Frontiers in Neurobotics*, 16, 846219.

A Supplementary Results

Appendix overview. The appendix is organized by how directly each result supports the main mechanism. The first block collects rule and feedback controls that complement Fig. 5. The second block reports competence checks, gradient dynamics, stress tests, and baseline comparisons. The third block contains harder-data, morphology, inhibition, and pathway-feedback diagnostics. The final blocks give implementation, theory, task, and concise secondary weight statistics. Exploratory screens and development summaries that do not directly support a paper claim are omitted.

Boundary diagnostics. Several controls are intentionally negative or mixed. The log-gain covariance margin from Eq. (8) is not a positive main result, Double-CIFAR remains near chance even under matched backpropagation, positive cue-routing controls saturate too strongly to separate mechanisms, and the current pathway-vector implementation does not beat the best unstructured low-rank cue-routing control. We include these results to delimit the claim rather than to smooth over failure modes.

Mechanistic Support and Broadcast Controls

Rule-Family Ranking

Dataset	Rule	Top-10 valid	Top-10 test
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	0.912	0.916
Context gating	3F	0.398	0.396
Context gating	4F	0.411	0.411
Context gating	5F	0.807	0.789

Table S1: **Rule-family ranking.** Top-10 mean across completed local-competence sweeps.

Competence, Verification, and Stress Tests

Local Competence and Regime Dependence

Extended Gradient Analysis and N_I Sweep Detail

Alignment dynamics are not a zero-gradient artifact. To check whether low additive alignment could be caused by vanishing gradients, we analyzed a separate 3-seed checkpoint trajectory with parameter-level LocalCA and backprop gradients. The additive local gradients are nonzero throughout training (weighted norm range 9.5×10^{-4} – 9.5×10^{-3}), and the corresponding backprop gradients are also nonzero (2.3×10^{-5} – 5.7×10^{-3}). The low additive cosine therefore reflects directional and scale mismatch, not absent gradients: additive 5F cosine stays near zero from epoch 0 to epoch 50 (0.018 to 0.005), while its local/backprop norm ratio falls from about 8.8×10^2 to 0.69.

Verification and Reproducibility

Additional Stress Tests

FA/DFA Baseline Comparison

CIFAR-10 Results

Table S3 provides the exact values for the appendix CIFAR-10 figure. In this stronger compact direct-I-stream family, shunting performs well under standard training (49.5% vs. 48.3% for additive), so the harder-data stress test does not unfairly penalize the shunting architecture. The broadcast-fidelity ladder also survives once decoder-aware soma mapping is used: additive LocalCA reaches 25.5% under rank-1 per-soma broadcast and 32.4% under transported error, while shunting rises from 29.7% under rank-1 broadcast to 35.0% with random low-rank $K=4$ and 47.2% with transported error,

Dataset	BP ceiling	Shunting local	Additive local	BP gap
MNIST	0.971	0.912 ± 0.005	0.894 ± 0.007	5.9%
Fashion-MNIST	0.889	0.838 ± 0.003	0.831 ± 0.004	5.1%
Context gating	0.861	0.803 ± 0.006	0.787 ± 0.007	5.8%

Table S2: **Local competence.** Backprop ceilings are matched-capacity shunting references from dedicated 5-seed standard-training refreshes for MNIST and context gating and from the dedicated 5-seed Fashion-MNIST competence sweep. Local values are 5F with rank-1 broadcast and local decoder updates. Context gating additionally uses an HSIC auxiliary objective (weight 0.01; Appendix C). Error bars on local values: ± 1 s.d. across 5 seeds.

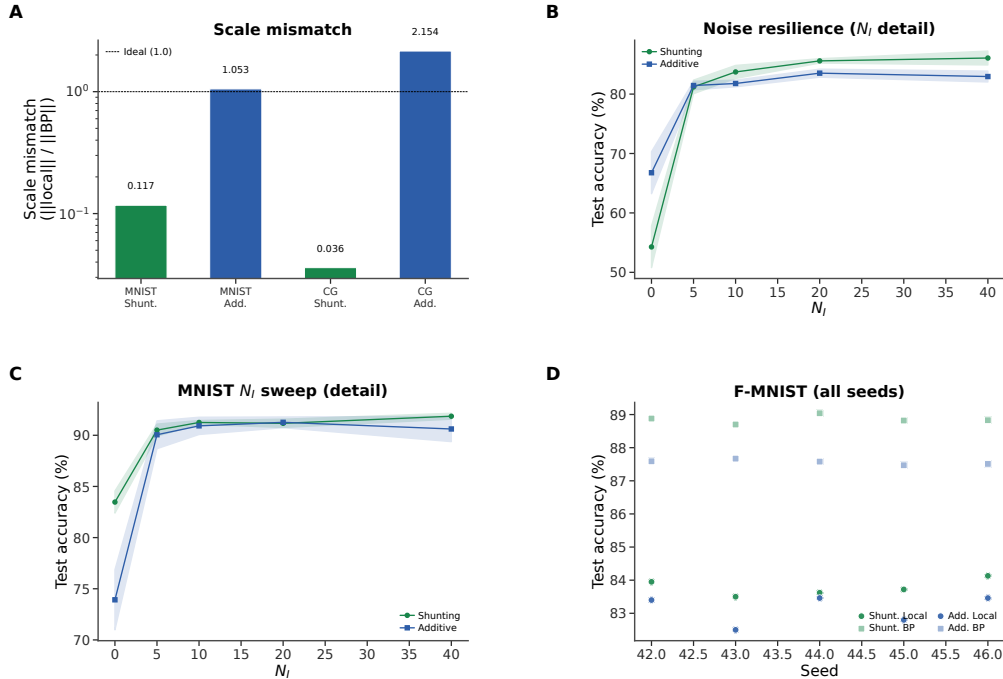


Figure S1: **Extended gradient and N_I analysis (supplementary).** (A) Scale mismatch bars from a separate matched-weight static diagnostic; these values should not be compared directly to the checkpointed final-state diagnostic in Fig. 2. Shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion (> 1.0). (B) Noise-resilience N_I dose-response with error bands (± 1 s.d.). (C) MNIST N_I dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

441 nearly matching the 49.5% shunting backprop ceiling. Removing learned I-to-E conductance lowers
 442 the shunting ceiling by 5.7 pp and transported LocalCA by 8.6 pp, making this the clearest harder-data
 443 evidence that inhibitory conductance is part of the useful credit geometry rather than just an added
 444 architectural detail.

445 A routed Double-CIFAR follow-up with strictly nonnegative router outputs was also not included as
 446 positive evidence. The task remained near chance even under matched backprop: additive BP reached
 447 $10.9 \pm 0.7\%$, shunting BP with learned I-to-E reached $11.6 \pm 0.4\%$, and the strongest LocalCA
 448 condition, shunting transported error with learned I-to-E, reached only $11.7 \pm 0.3\%$. We therefore
 449 treat this as a failed hard routed-task screen and keep the stronger compact CIFAR-10 control ladder
 450 as the paper’s harder-data diagnostic.

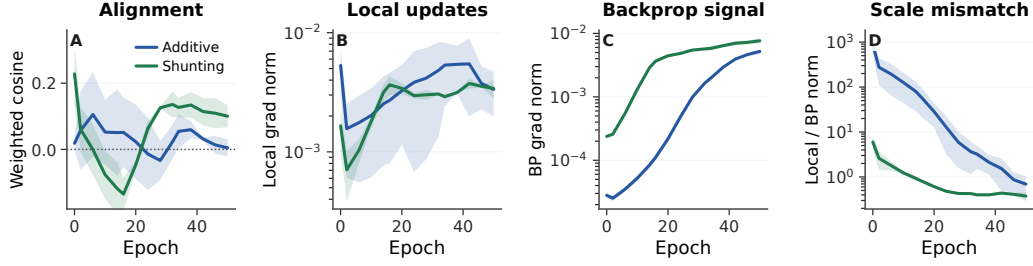


Figure S2: **Alignment dynamics with explicit gradient-norm checks.** (A) Weighted LocalCA-backprop cosine over training for the 5F rule. (B,C) Weighted local and backprop gradient norms stay nonzero for both additive and shunting models. (D) The additive control begins with a much larger scale mismatch, then approaches the backprop scale without recovering strong directional alignment. Shading is s.d. across 3 seeds.

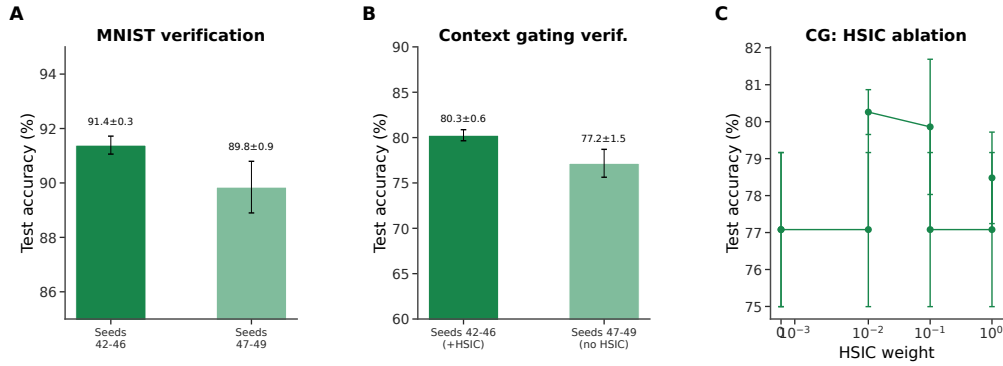


Figure S3: **Verification and reproducibility (supplementary).** (A) MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. (B) Context gating verification: main seeds (+HSIC) $80.3 \pm 0.6\%$; held-out seeds (no HSIC) $77.2 \pm 1.5\%$. The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. (C) HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

451 Additive Normalization Control

452 Dendritic Morphology, Pathway Structure, and Supportive Extensions

453 Morphology $\times$ Inhibition Regime Map

454 This supportive regime map tests whether tree geometry changes the inhibitory operating point at
 455 which shunting helps. We treat it as evidence that morphology changes the useful inhibition regime;
 456 a fuller mechanistic account would still require matched theory diagnostics for each cell in the grid.

457 Input-Mode Probe: Direct Inhibitory Stream vs. Explicit Inhibitory Cells

458 The main experiments use input-driven inhibitory streams: the transfer layer provides nonnegative
 459 excitatory and inhibitory channels from the external input, and excitatory dendrites receive learned
 460 I-to-E conductance banks directly. To test whether the same mechanism can be generated by an
 461 explicit feedforward inhibitory population, we ran a one-feedforward-layer noise-resilience probe
 462 that preserves the relevant dendritic depth ([3, 3] morphology). In this clean setting, explicit inhibitory
 463 cells reach $94.2 \pm 0.2\%$ under path-transport LocalCA when their dendrites use the same update policy
 464 as excitatory dendrites, $93.8 \pm 0.2\%$ when those inhibitory-cell updates are frozen, and $95.1 \pm 0.3\%$
 465 under standard backprop. Direct input-driven inhibition gives the expected broadcast-fidelity ladder
 466 on the same one-layer architecture: rank-1 broadcast reaches $85.2 \pm 0.7\%$, while exact path transport
 467 reaches $95.2 \pm 0.0\%$. Thus explicit inhibitory cells can generate useful path-gain structure once
 468 the architecture isolates the dendritic-path question. The older two-feedforward-layer explicit-I
 469 diagnostic was much weaker (about 58–63% under LocalCA despite a 94.7% backprop ceiling), so

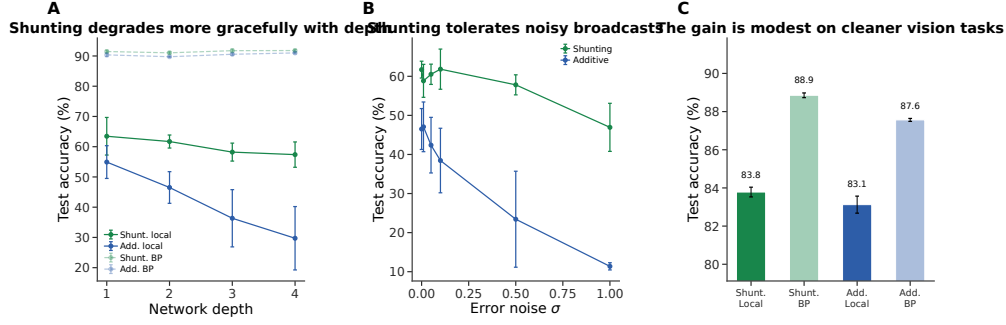


Figure S4: **Additional stress tests.** (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent story in the main text.

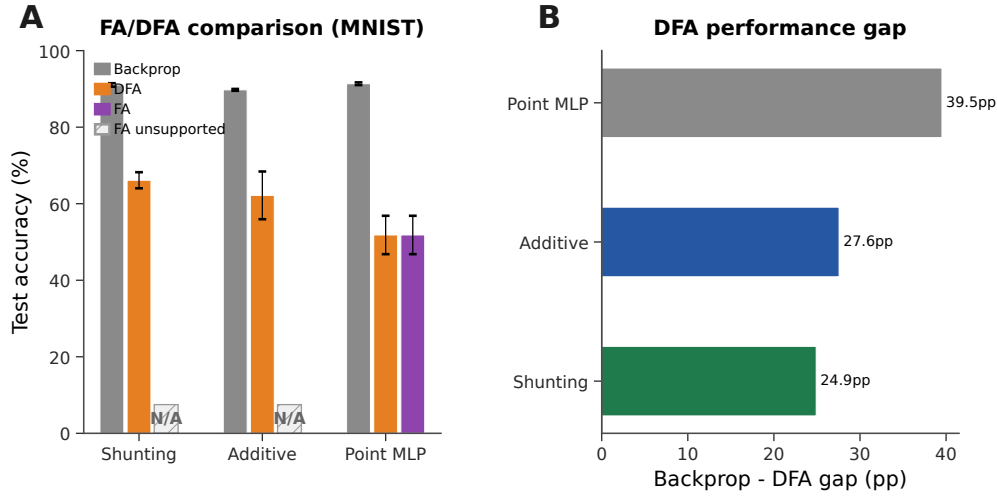


Figure S5: **Feedback alignment baselines (MNIST).** (A) Grouped comparison: standard backprop (neutral gray), DFA (orange), and FA (purple). Hatched N/A markers indicate that FA is not implemented for the block-structured dendritic architectures because the random feedback matrices are not dimensionally compatible with the tree-structured branch updates. DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. (B) Backprop–DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

470 we treat it as evidence that deeper explicit $E \rightarrow I \rightarrow E$ graphs need graph-aware local errors, not as
 471 evidence against inhibition-mediated path gains.

472 **Post-training inhibition intervention and exact-error rank.** We next tested whether learned
 473 inhibitory conductance carries task-specific structure rather than only changing global scale. We
 474 replayed trained shunting LocalCA checkpoints while replacing the branch-level inhibitory conduc-
 475 tance with four post-training interventions: zero inhibition, sample-shuffled inhibition, per-branch
 476 batch means, or one uniform matched mean. On a 512-example held-out subset, learned MNIST
 477 inhibition at $N_I=5$ gives $90.4 \pm 1.1\%$ accuracy, while the four interventions drop to 19.8–26.4%.
 478 On noise resilience at $N_I=10$, learned inhibition gives $80.5 \pm 2.4\%$, while the interventions drop
 479 to 10.3–19.3%. Thus the inhibitory field is not interchangeable with a matched global conductance
 480 load. We also computed the SVD of exact compartment-error matrices across samples and compart-
 481 ments. At matched MNIST $N_I=5$, shunting has lower rank-1 residual than additive (0.65 ± 0.18

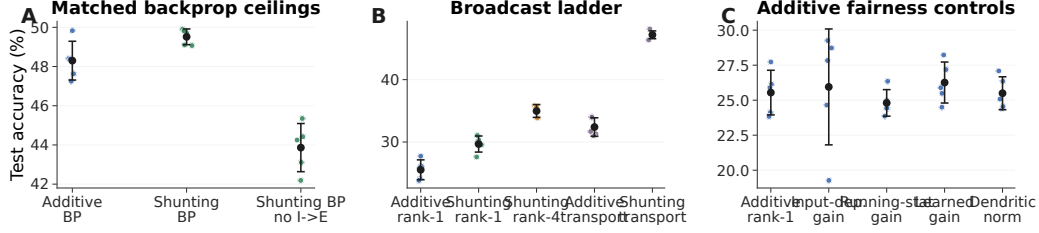


Figure S6: **CIFAR-10 control ladder in the compact direct-I-stream family.** (A) Matched backprop ceilings. With nonnegative inputs and a nonlinear decoder, shunting slightly exceeds additive under standard training, and removing learned I-to-E conductance lowers the shunting ceiling. (B) Broadcast ladder. Rank-1 broadcast remains weak on this harder dataset, rank-4 improves partially, and exact transported error nearly reaches the shunting backprop ceiling. (C) Additive fairness controls. Input-dependent gain, running-stat gain, learned gain, and dendritic normalization do not rescue additive rank-1 broadcast. This figure is a harder-data diagnostic, not a claim of competitive CIFAR-10 benchmarking.

Condition	Core	Training / broadcast	I-to-E	Test acc.
standard	additive	backprop	yes	48.3 ± 1.0
standard	shunting	backprop	yes	49.5 ± 0.4
standard, no learned inhibition	shunting	backprop	no	43.9 ± 1.2
rank-1	additive	5F per-soma	yes	25.5 ± 1.6
rank-1, input-dependent gain	additive	5F per-soma	yes	25.9 ± 4.1
rank-1, running-stat gain	additive	5F per-soma	yes	24.8 ± 0.9
rank-1, learned gain	additive	5F per-soma	yes	26.3 ± 1.5
rank-1, dendritic normalization	additive	5F per-soma	yes	25.5 ± 1.2
rank-1	shunting	5F per-soma	yes	29.7 ± 1.3
rank-1, no learned inhibition	shunting	5F per-soma	no	22.3 ± 0.7
transported error	additive	5F path transport	yes	32.4 ± 1.5
rank-4	shunting	5F low-rank	yes	35.0 ± 1.0
transported error	shunting	5F path transport	yes	47.2 ± 0.6
transported error, no learned inhibition	shunting	5F path transport	no	38.6 ± 1.0

Table S3: **Exact CIFAR-10 values for the 70-run control ladder (5 seeds per condition, validation-best epoch).** All rows use flattened CIFAR-10 in $[0, 1]$, nonnegative transfer outputs, positive conductance transforms, a compact depth-4 dendritic core, and a nonlinear decoder. LocalCA maps output error back to decoder-input/soma coordinates through the decoder Jacobian. The useful conclusions are narrow but important: learned shunting slightly improves the matched backprop ceiling, learned I-to-E conductance is necessary for the strongest shunting results, transported error nearly closes the shunting LocalCA gap, and additive rank-1 fairness controls do not explain the transported-shunting result.

482 vs. 0.83 ± 0.07) and lower participation rank (3.2 ± 1.2 vs. 8.5 ± 3.3), supporting the path-gain
483 compressibility interpretation.

484 We also computed the direct log-gain covariance margin in Eq. (8) on the selected morphology
485 diagnostic checkpoints. The margin is zero when no inhibitory conductance is present and slightly
486 negative in the inhibited shunting selections (-2.5×10^{-3} , -5.0×10^{-4} , and -2.8×10^{-4} at
487 $N_I=5, 20, 40$). We therefore do not use this covariance condition as a positive main result. Its
488 role is to clarify when shunting should compress gains; the empirical claim rests on the measured
489 comparative path-gain dispersion, exact-error rank, and inhibition-intervention diagnostics.

490 Cue-Routing Feedback-Rank Diagnostic

491 Cue routing tests the regime where one shared teaching signal is expected to fail: context determines
492 which noisy cue is reliable, so different pathways should receive different credit. Figure S10 shows
493 that the benchmark is learnable and that moving beyond rank-1 feedback helps, but the current
494 path-structured variant does not yet beat the best unstructured low-rank setting. We therefore interpret
495 this result as a failure-mode and rank-requirement diagnostic, not as a solved structured-feedback
496 method.

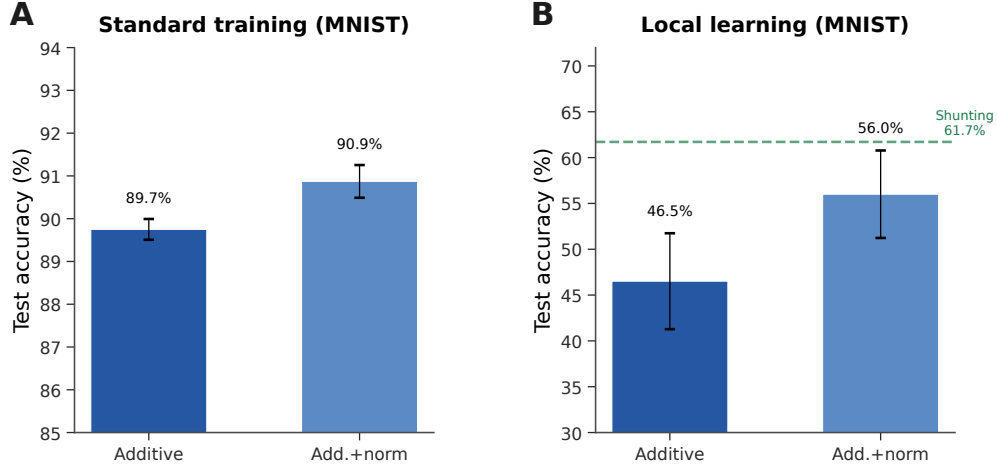


Figure S7: **Additive + normalization control (MNIST).** (A) Standard backprop: normalization provides a small boost (89.7% $\rightarrow$ 90.9%). (B) Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). The remaining gap (−5.7 pp) indicates that shunting provides benefits beyond simple voltage normalization.

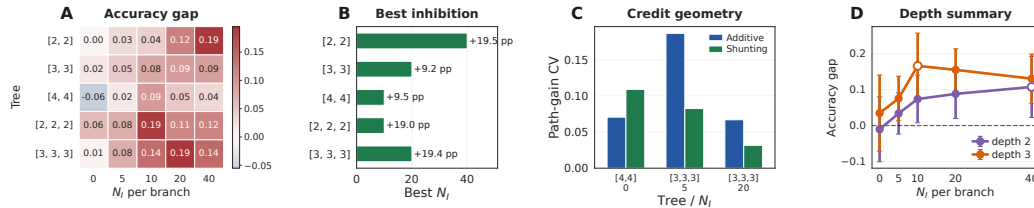


Figure S8: **Morphology shifts the useful inhibitory regime.** (A) Shunting-minus-additive LocalCA accuracy gap across dendritic morphologies and inhibitory synapse counts. (B) Best inhibitory count for each morphology. (C) Representative path-gain diagnostic for selected morphology cells, used to connect the performance map to credit geometry. (D) Average shunting advantage by depth, with markers showing the peak inhibitory count for each depth.

Random Low-Rank Broadcast Channels

Frozen Activation-Choice Audit

B Implementation Details

Biological Plausibility Assumptions

The theorems and rules above rest on a small set of explicit assumptions. We state them as a checklist so that readers can see exactly what is assumed and where each assumption is used:

A1 (Steady-state voltage). We use the steady-state solution of the passive cable equation (Eq. (1)).

Temporal dynamics are not modeled; all gradients, path gains, and fidelity measurements refer to the conductance-stage voltage at steady state.

A2 (Tree topology). Each dendritic cell is a rooted tree with the soma as the root. There are no recurrent connections within the dendrite, which makes the path from any compartment to the soma unique and is what allows the closed-form path gain α_n in Theorem 1.

A3 (Nonnegative conductances). Synaptic and dendritic conductance parameters are pushed through a positive transform (`softplus`), so $g_j^{\text{syn}}, g_j^{\text{den}} \geq 0$ at every optimisation step. This is a biophysical constraint and is also what keeps the shunting denominator bounded away from zero.

Inhibitory input mode	Training / broadcast	Test accuracy
Direct inhibitory stream (input_mode=1)	LocalCA, per-soma rank-1	0.852 ± 0.007
Direct inhibitory stream (input_mode=1)	LocalCA, exact path transport	0.952 ± 0.000
Explicit I cells (input_mode=0)	LocalCA, path transport, I-cell updates on	0.942 ± 0.002
Explicit I cells (input_mode=0)	LocalCA, path transport, I-cell updates frozen	0.938 ± 0.002
Explicit I cells (input_mode=0)	standard backprop	0.951 ± 0.003

Table S4: **One-feedforward-layer input-mode path-gain probe on noise resilience (3 seeds).** All rows use one excitatory dendritic population with [3, 3] morphology and nonnegative inputs. The direct-I rows have no explicit inhibitory neurons; the input stream drives learned branch-level inhibitory conductance banks directly. The explicit-I rows use a matched inhibitory population with [3, 3] morphology. `local_ca` updates explicit inhibitory-cell dendrites with the same policy used for excitatory dendrites, while `freeze` holds those dendrites fixed but still lets excitatory I-to-E synapses learn. Exact path transport closes the direct-I rank-1 gap, and explicit inhibitory cells nearly match both direct-I path transport and the explicit-I backprop ceiling.

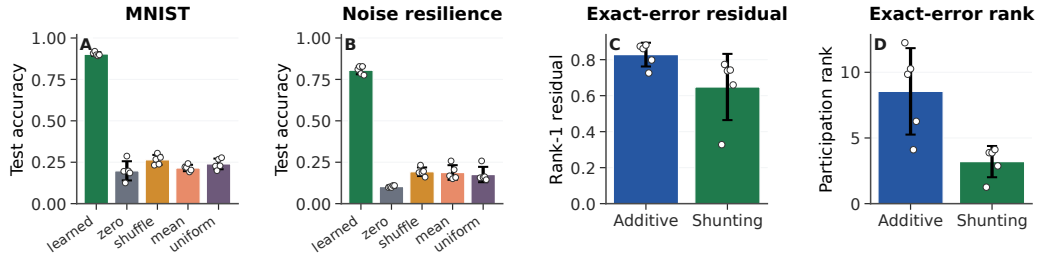


Figure S9: **Post-training inhibition intervention and exact-error compressibility.** (A,B) Post-training inhibitory-conductance interventions on trained shunting LocalCA checkpoints. Accuracy is measured on a 512-example held-out subset; dots show seeds. Removing, shuffling, or clamping learned inhibition collapses performance, showing that the learned inhibitory field carries sample-specific task structure. (C,D) SVD diagnostics of the exact compartment-error matrix on matched MNIST $N_I=5$ runs. Shunting has a smaller rank-1 residual and lower participation rank than additive, so the exact error field is more compressible. Error bars are s.d. across 5 seeds.

513 **A4 (Nonnegative presynaptic drive).** The first synaptic drive x is nonnegative, implemented either
 514 by nonnegative inputs or an explicit ReLU transfer stage. Together with A3 this enforces
 515 the positive-input condition used for all main sweeps.

516 **A5 (Pre-reactivation upward transfer).** Theorem 1 is stated for the pre-reactivation compartment
 517 voltage. A post-voltage reactivation f (e.g. `param_tanh`) is handled by multiplying the
 518 path gain by $f'(V_n)$ along the recursion. The compartment error remains the sole non-local
 519 quantity; only the effective path gain acquires local activation-derivative factors.

520 **A6 (Single perisomatic broadcast).** Every synapse receives one non-local scalar or low-rank vector
 521 signal delivered from the soma; all other ingredients of the update rule (eligibility traces,
 522 driving force, input resistance, covariance modulators) are computable from quantities
 523 available at the synapse or compartment.

524 Each synapse therefore has access to: presynaptic activity x_j (local), compartment voltage V_n
 525 (local membrane potential), reversal potential E_j (fixed by receptor type), and total input resistance
 526 $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ (computable from local conductances). The 4F modulator r_n^{4F} requires online
 527 estimation of voltage covariance (biologically plausible via slow calcium signals); the 5F factor ϕ_n
 528 requires a linear regression proxy (implementable via eligibility traces). The only non-local quantity
 529 is the broadcast error e_n , which requires a top-down or neuromodulatory signal from the soma to
 530 dendritic compartments. We assume per-neuron resolution (one error per output neuron) for the
 531 per-soma broadcast mode.

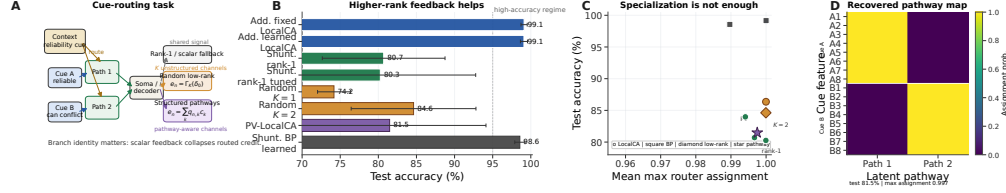


Figure S10: **Structured feedback when rank-1 broadcast fails.** (A) Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. (B) Five-seed comparison of rank-1, random low-rank, and pathway-structured feedback. Moving beyond rank-1 helps, while the best structured repair remains open. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. (D) Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate into separate latent pathways even when feedback quality, rather than router discreteness, is the limiting factor.

Setting	Broadcast	Test accuracy
<i>Noise resilience, shunting, rank bridge</i>		
per-soma	rank-1 per-soma	0.662 ± 0.016
path propagation	recursive attenuation proxy	0.762 ± 0.047
low-rank	$K=1$	0.453 ± 0.126
low-rank	$K=2$	0.630 ± 0.042
low-rank	$K=4$	0.764 ± 0.025
low-rank	$K=8$	0.795 ± 0.019
path transport	exact conductance-stage transport	0.847 ± 0.018
<i>Cue routing, learned shunting router, rank/structure sweep</i>		
per-soma	tuned rank-1	0.803 ± 0.125
low-rank	$K=1$	0.742 ± 0.022
low-rank	$K=2$	0.846 ± 0.082
low-rank	$K=4$	0.748 ± 0.085
pathway-vector	tuned structured rank-2	0.815 ± 0.126

Table S5: **Rank bridge and rank-structure comparison.** On shunting noise resilience, both the current path-propagation proxy and higher-rank random low-rank broadcast close a substantial fraction of the gap between rank-1 per-soma and exact path transport, with low-rank continuing to improve up to $K=8$. On cue routing, the learned-router sweep shows that moving beyond rank-1 helps, but the best current unstructured low-rank setting ($K=2$) is still slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

532 Dendritic Structure in Code

533 We use the morphology notation $[b_1, b_2, \dots, b_D]$ for rooted dendritic trees, where each factor gives
534 the fan-in from one dendritic stage to the next on the path to the soma. For example, $[3, 3]$ means that
535 each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches,
536 for 9 distal leaves per soma. In the main feedforward models, synaptic inputs terminate on dendritic
537 branches rather than directly on the soma. The main setting uses direct input-driven inhibition: the
538 transfer layer creates nonnegative excitatory and inhibitory streams from the same external input, and
539 learned inhibitory conductance banks attach directly to excitatory dendritic branches. These models
540 therefore have branch-level inhibitory conductance but no separate inhibitory neuron population. The
541 explicit-I probe instead instantiates feedforward inhibitory cells, which transform excitatory activity
542 before projecting inhibitory conductance back onto excitatory dendrites. In that setting the inhibitory
543 cells are also dendritic modules and their synaptic conductances are trained by the same local rule
544 family. The additive and shunting cores are architecture-matched at the tree level; they differ in the
545 branch-level voltage integration rule rather than in whether a dendritic tree exists. In all reported
546 dendritic runs, synaptic and dendritic conductance parameters are pushed through positive transforms,

Dataset / strategy	Core	N_I	Identity (none)	Frozen param_tanh
MNIST / LocalCA	additive	0	0.780 ± 0.014	0.709 ± 0.037
MNIST / LocalCA	additive	10	0.907 ± 0.003	0.881 ± 0.005
MNIST / LocalCA	shunting	0	0.841 ± 0.007	0.778 ± 0.020
MNIST / LocalCA	shunting	10	0.905 ± 0.009	0.840 ± 0.016
Noise resilience / LocalCA	additive	0	0.590 ± 0.162	0.109 ± 0.006
Noise resilience / LocalCA	additive	10	0.859 ± 0.001	0.772 ± 0.009
Noise resilience / LocalCA	shunting	0	0.411 ± 0.028	0.331 ± 0.020
Noise resilience / LocalCA	shunting	10	0.769 ± 0.004	0.701 ± 0.006
MNIST / backprop	additive	0	0.925 ± 0.001	0.942 ± 0.001
MNIST / backprop	additive	10	0.921 ± 0.001	0.950 ± 0.001
MNIST / backprop	shunting	0	0.963 ± 0.0004	0.950 ± 0.001
MNIST / backprop	shunting	10	0.967 ± 0.001	0.946 ± 0.001
Noise resilience / backprop	additive	0	0.896 ± 0.003	0.106 ± 0.007
Noise resilience / backprop	additive	10	0.893 ± 0.002	0.927 ± 0.002
Noise resilience / backprop	shunting	0	0.898 ± 0.001	0.869 ± 0.002
Noise resilience / backprop	shunting	10	0.926 ± 0.001	0.912 ± 0.001

Table S6: **Frozen activation-shape audit on matched dendritic architectures (5 seeds per condition).** Identity (none) and a frozen bounded post-voltage reactivation param_tanh lead to markedly different operating regimes under nonnegative first-layer synaptic drive. In this audit, identity is the safer LocalCA choice on both MNIST and the noise-resilience task, while the backprop effect is strongly regime-dependent: frozen param_tanh helps additive most clearly in the inhibited noise-resilience setting ($N_I=10$) but collapses it at $N_I=0$. These runs match additive reactivation initialization to the shunting setting and hold the activation parameters fixed, isolating activation shape from both additive/shunting initialization differences and activation learning itself.

Dataset	Core	N_I	Identity (none)	Frozen param_tanh
MNIST	additive	0	CV 0.280, cosine 0.220	CV 1.122, cosine 0.074
MNIST	additive	10	CV 0.310, cosine 0.212	CV 1.079, cosine 0.094
MNIST	shunting	0	CV 0.393, cosine 0.170	CV 0.770, cosine 0.114
MNIST	shunting	10	CV 0.385, cosine 0.281	CV 1.177, cosine 0.106
Noise resilience	additive	0	CV 0.124, cosine 0.079	CV 0.114, cosine 0.052
Noise resilience	additive	10	CV 0.414, cosine 0.206	CV 0.840, cosine 0.094
Noise resilience	shunting	0	CV 0.255, cosine 0.324	CV 0.344, cosine 0.341
Noise resilience	shunting	10	CV 0.762, cosine 0.140	CV 2.497, cosine 0.079

Table S7: **Frozen activation-shape theory audit on matched LocalCA runs (5 seeds per condition).** Each entry reports conductance-stage path-gain dispersion (CV) and direct per-soma compartment-error fidelity (cosine between the per-soma broadcast and the exact pre-reactivation compartment error). The positive-input theory audit largely agrees with the empirical one: freezing param_tanh usually broadens path-gain dispersion and weakens per-soma fidelity, especially in the inhibited settings where the main mechanistic story lives. The one mild exception is low-inhibition shunting on noise resilience, where frozen param_tanh leaves fidelity roughly unchanged while still hurting final accuracy, suggesting that activation choice affects optimization and representation as well as direct credit fidelity.

547 so conductances remain nonnegative; unconstrained decoder weights are separate readout parameters
548 and are not interpreted as conductances.

549 Post-Voltage Reactivation

550 The conductance stage produces a pre-reactivation compartment voltage V_n , after which each branch
551 layer may apply a monotone scalar nonlinearity. The main sweeps use a learnable bounded firing-rate
552 transform, $(\tanh(m(V - b)) + 1)/2$, together with a nonnegative transfer stage so that the first
553 synaptic drive remains nonnegative even after input corruption or structured task noise. Most main

Quantity	Symbol	Convention
Voltage	V	Conductance-stage shunting voltage in $[0, 1]$ under nonnegative drive; additive controls may be signed
Conductances	$g^{\text{syn}}, g^{\text{den}}$	Nonneg. via softplus
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S8: Units and normalization.

feedforward sweeps use a linear decoder, while the CIFAR extension uses a nonlinear decoder and therefore maps output error back to decoder-input/soma space through the decoder Jacobian before applying LocalCA. Unless explicitly overridden, additive and shunting cores can start from different reactivation initializations; activation-fairness ablations therefore match additive initialization to the shunting setting when comparing alternative reactivation choices directly. The appendix activation audit additionally fixes the bounded reactivation after initialization to isolate activation shape from activation plasticity.

LocalCA uses occupancy-quantile initialization together with a calibration pass that floors tiny quantile widths, clamps extreme fitted slopes, and reverts invalid per-layer fits rather than allowing pathological intermediate values to propagate. In the non-somatic feedforward publication families, this calibration leaves the results essentially unchanged relative to the same occupancy-quantile setup without the additional validity checks: phase-1 capacity changes from 0.8786 to 0.8784 mean test accuracy, shunting-regime sweeps from 0.5617 to 0.5641, morphology-scaling sweeps from 0.3600 to 0.3578, and error-shaping sweeps from 0.2860 to 0.2798. The calibration therefore improves numerical safety without changing the qualitative LocalCA story.

Component Checks

A five-seed MNIST component check confirmed that the main LocalCA results are not driven by a single calibration artifact. Removing the occupancy-quantile calibration changes shunting accuracy negligibly (90.7% to 90.8%), removing learned reactivation parameters costs about 1 pp, and removing the post-voltage reactivation is the largest single ablation (87.3%). Additive controls show small gains from each calibration ingredient. We keep these checks as implementation support rather than as a separate appendix figure.

Units and Parameterization

Decoder Update Modes

W_{dec} maps $V_L \rightarrow \hat{y}$. Modes: **backprop** ($\nabla_W L$ via autograd), **local** ($\Delta W = \eta \langle \delta_0 V_L^\top \rangle_B$), **frozen** ($\Delta W = 0$).

Representative Manuscript Architectures

Hyperparameters

Table S10 collects the core training hyperparameters for each main experiment. Learning rates are applied through parameter groups (top- k , dendritic block-linear, reactivation, decoder) with the values given in the representative config files; where a single LR is reported, every group uses that value.

Effective Broadcast Dimensionality in Main Architectures

Compute Resources

All experiments were run on an institutional GPU cluster. Most MNIST, Fashion-MNIST, and synthetic-task runs used a single NVIDIA A100 or H100 GPU and finished in minutes to tens of minutes per seed. The flattened CIFAR-10 runs required longer single-GPU jobs but still did not use distributed or multi-GPU training. Reported main metrics are aggregated over 5 seeds unless noted

otherwise; selected morphology and input-mode probes use 3 seeds and are labeled as supportive diagnostics.

Code, Data, and Asset Availability

MNIST, Fashion-MNIST, CIFAR-10, and MICrONS are public datasets or public benchmark assets cited in the paper; we use them under their standard published access conditions and cite their original sources in the bibliography. To preserve double-blind review, the submission provides code through an anonymized supplementary ZIP rather than a public repository URL. The ZIP contains the model, training, diagnostic, and figure-generation code together with representative configuration files and reproduction instructions for the main experimental families; code and configurations will be de-anonymized after review. The paper does not release a new dataset or a stand-alone pretrained model asset.

Algorithm

Algorithm 1 Main 5F Rank-1 LocalCA Update

- 1: **Input:** Dendritic model, batch (x, y) , learning rate η
 - 2: Forward pass; loss L , output error δ^y
 - 3: Somatic/core error $\delta_0 = \text{decoder-input Jacobian product } J_{\text{dec}}(h_{\text{core}})^\top \delta^y$ (or $W_{\text{dec}}^\top \delta^y$ for a linear decoder)
 - 4: **for** each layer n (reverse) **do**
 - 5: Rank-1 broadcast $e_n = \text{mean}(\delta_0) \mathbf{1}_n$
 - 6: Update EMA estimates for r_n^{4F} and ϕ_n from batch voltages
 - 7: $\Delta g_j^{\text{syn}} \leftarrow \eta r_n^{4F} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B$
 - 8: $\Delta g_j^{\text{den}} \leftarrow \eta r_n^{4F} \phi_n \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B$
 - 9: **end for**
 - 10: Optional controls replace line 4 with neuron-wise, rank- K , path-structured, or transported-oracle broadcasts.
 - 11: Clip gradients; optimizer step
-

C Theoretical Details

Random-Broadcast Alignment

Proposition 3 (Supportive random-broadcast alignment). *Let the broadcast matrix B_n have i.i.d. zero-mean entries with $\mathbb{E}[B_n^\top B_n] = \sigma_B^2 I$. Then the expected cosine between the resulting local gradient and the exact gradient is positive, $\mathbb{E}[\cos \angle(g^{\text{local}}, g^{\text{exact}})] \geq c_n > 0$, by an argument analogous to feedback alignment [9]. The constant c_n depends on how strongly local eligibility factors correlate with the conductance-stage path gain in Eq. (4); shunting improves this correlation by narrowing the spread of intermediate sensitivities.*

We treat this argument as supportive rather than central. The main theoretical object in the paper is still the exact pre-reactivation compartment error $\partial L / \partial V_n$; under identity upward transfer this reduces to $\alpha_n \delta_0$, while with post-voltage activation the corresponding effective path gain additionally includes local $f'_n(V_n)$ factors. The main empirical question is how faithfully different broadcast fields approximate that quantity.

Morphology-Aware Extensions

Path-integrated propagation. Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approximating depth attenuation from Eq. (4).

Depth modulation. Per-branch scaling $\kappa_j = \kappa_{\text{base}} / (d_j + d_0)$, mirroring cable attenuation.

Dendritic normalization. $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}} / (\sum_k g_k^{\text{den}} + \varepsilon)$, analogous to homeostatic scaling [21].

Pathway-vector feedback. For tasks with latent pathway structure, the broadcast becomes a role vector inferred from the router. Local pathway activity gates this vector. Upward block transport then passes it to earlier branches, aligning their feedback with downstream descendants.

Router-derived branch roles. Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

HSIC Auxiliary Objectives

Following [17], we use kernel-based HSIC objectives:

$$\mathcal{L}^{\text{self}} = B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H}), \quad \mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H}).$$

Moderate weights (0.01–0.1) help on context gating; negligible on MNIST. Online statistics (r_n^{4F} , ϕ_n) use Welford’s algorithm [20].

Online Variant with Eligibility Traces

The continuous-time eligibility trace is

$$\tau_e \dot{e}_j^{\text{syn}} = -e_j^{\text{syn}} + x_j(E_j - V_n)R_n^{\text{tot}},$$

with update

$$\Delta g_j^{\text{syn}} \propto \int e_j^{\text{syn}}(t) m_n(t) dt$$

[18, 19].

D Secondary Weight Statistics

As a secondary biological check, we fit log-normal distributions to active excitatory and inhibitory synaptic weights after training on MNIST and Fashion-MNIST. This analysis is descriptive rather than central to the credit-assignment claim. The main qualitative pattern is that shunting produces narrower and more stable weight distributions than additive integration, with the shunting conditions falling closer to descriptive biological reference widths from Song et al. [38] and MICrONS [40].

The same self-normalizing denominator that shapes credit also provides a plausible explanation for this secondary trend: larger total conductance lowers R_n^{tot} , reducing future update magnitudes on already high-conductance branches. This resembles multiplicative synaptic dynamics [41], but we do not use weight-distribution fits as evidence for the main credit-assignment claims.

E Task and Robustness Details

Each synthetic task is generated procedurally from a fixed random seed and uses fixed train/val/test splits for reproducibility. The MNIST-derived synthetic tasks use flattened 28×28 inputs; cue integration instead uses a small vector input with two output classes. Output dimensionality therefore varies by task and is stated in the task description when it differs from 10-way digit classification.

Context gating. Inputs are contextual figure-ground MNIST examples. The right half of the image carries the original MNIST digit, while the left half is replaced by low-amplitude uniform noise; all pixels therefore remain nonnegative. This task tests whether credit assignment can exploit a simple context-dependent spatial structure without leaving the nonnegative-drive regime of the conductance model. The HSIC auxiliary objective (weight 0.01) is used to decorrelate layer representations, adding roughly 3 pp under local learning.

Noise resilience. Inputs are flattened MNIST images corrupted by structured channel noise: a fixed 50-dimensional latent Gaussian perturbation is projected into input space and added to each example, with $\sigma_{\text{task}} = 1.5$, then clipped to $[0, 1]$. The projection is fixed across training, so the network must filter correlated interference to recover the underlying class signal. Credit assignment must remain coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

663 **Info shunting.** Inputs contain two additive streams: a low-frequency signal and a high-frequency
664 distractor drawn with equal energy, together with a separate context channel indicating which stream
665 is relevant on each trial. This benchmark uses signed synthetic inputs, so it violates the nonnegative
666 presynaptic-drive assumption enforced in the main experiments. We therefore retain it only as an
667 exploratory stress test and exclude it from the main-text claims.

668 **Cue integration.** Two noisy cue streams (A and B) are presented simultaneously, each carrying
669 partial information about one of 2 classes. A binary one-hot context signal indicates which cue is
670 reliable on each trial (cue A or cue B), and each cue vector is clipped to $[0, 1]$ after noise is added.
671 The fixed-pathway variant duplicates the context into both cue branches; the learned-routing variant
672 requires the router to discover the cue separation from data. This task is used in Sec. A to test
673 structured pathway-vector broadcast while staying inside the nonnegative-input regime.

674 **Depth Scaling and Noise Robustness**

675 **Depth scaling.** Varying dendritic depth from 1–4 layers (branch factors $[9]$ to $[3, 3, 3, 3]$): shunting
676 local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%.
677 The shunting advantage grows with depth ($+8.5 \rightarrow +27.7$ pp). Backprop ceilings remain stable at
678 about 90–92%. See Fig. S4A.

679 **Noise robustness.** Gaussian noise $\mathcal{N}(0, \sigma^2)$ on broadcast error: shunting is robust to $\sigma \leq 0.1$ (about
680 62%) while additive drops from 46.5% to chance at $\sigma=1.0$, confirming that shunting credit signals
681 carry useful learning information beyond the broadcast magnitude. See Fig. S4B.

Setting	Encoder	E layers	Tree	Synapses / branch	Seeds	Notes
Gradient fidelity / path-gain / noise-resilience mechanism	id.+ReLU	[128, 128]	[3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	5 fidelity, 5 oracle	Main mechanism sweeps in Figs. 2 and 3; reactivation <code>param_tanh</code> .
MNIST / context-gating local competence	id.+ReLU	[128]	[3, 3]	$N_E=40, N_I=20$	5	Best local 5F per-soma runs in Table S2; reactivation <code>param_tanh</code> .
Fashion-MNIST competence	id.+ReLU	[128]	[3, 3]	$N_E=40, N_I=20$	5	Dedicated five-seed standard-vs-local competence sweep; reactivation <code>param_tanh</code> .
Cue-routing PV-LocalCA	router	[64]	[2]	$N_E=10, N_I=4$	5	Learned router with two pathway groups and rank-2 structured feedback; reactivation <code>param_tanh</code> .
CIFAR-10 compact direct-I-stream control ladder	id.+ReLU	[20] E, no I cells	[3, 3, 3, 3]	$N_E=25, N_I=25$ direct I-to-E, with matched no-I controls	5 per condition	Harder-data extension with input-driven inhibitory conductance, decoder [32, 16], decoder-aware soma mapping, and additive fairness controls.
Morphology $\times$ inhibition appendix map	id.+ReLU	[128, 128]	[2, 2], [3, 3], [4, 4], [2, 2, 2], [3, 3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	3	Supportive 3-seed sweep; reactivation <code>param_tanh</code> .

Table S9: Representative architectures used in the manuscript. All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks with `somatic_synapses=false`, nonnegative conductances, and a nonnegative first-layer drive enforced by an explicit ReLU transfer stage in the main runs. The tree column gives dendritic morphology per excitatory neuron, and the synapse counts report excitatory and inhibitory synapses per branch.

Setting	Opt.	LR	Batch	Epochs	W-dec.	Notes
MNIST / Fashion-MNIST / context gating (5F per-soma LocalCA)	Adam	$10^{-3} / 5 \cdot 10^{-4}$ (block / react.)	256	100	0	fixed-epoch, no early stopping; <code>param_tanh</code> reactivation
MNIST / Fashion-MNIST / context gating (BP ceiling)	Adam	$10^{-3} / 5 \cdot 10^{-4}$	256	100	0	matched to LocalCA setup
Gradient fidelity / path-gain / noise resilience	Adam	10^{-3}	256	50	0	5 seeds, hooks enabled for diagnostic capture
Morphology $\times$ inhibition regime map (exploratory)	Adam	10^{-3}	256	50	0	3 seeds
Cue routing	Adam	10^{-3}	256	90	0	early stopping with patience 30
CIFAR-10 compact direct-I-stream depth-4 LocalCA	Adam	$10^{-3} / 10^{-4}$ (block / react.)	256	400	0 / 0.01	grad-clip 5.0, early stop patience 50, nonlinear decoder
CIFAR-10 compact direct-I-stream depth-4 BP	Adam	10^{-3}	256	200	0.01	early stop patience 40, matched ceiling
Weight-statistics summary	Adam	10^{-3}	256	100	0	<code>weight_analysis</code> hook captures final-epoch statistics

Table S10: Training hyperparameters by experiment. All runs use Adam with default $\beta_1=0.9$, $\beta_2=0.999$, $\varepsilon=10^{-8}$. Reactivation and block-linear parameter groups use a slower LR to keep conductance-level dynamics stable. Full configs are provided under `drafts/dendritic-local-learning/configs/`.

Setting	Hidden width(s)	Output dim	Effective default per-soma field
Gradient fidelity / path-gain / noise resilience	128, 128	10	Scalar / rank-1 at both dendritic layers
MNIST / Fashion-MNIST competence	128	10	Scalar / rank-1 at the dendritic hidden layer
Context gating competence	128	2	Scalar / rank-1 at the dendritic hidden layer
Cue routing	64	2	Scalar / rank-1 by default; higher-rank tests use explicit low-rank or pathway-vector feedback
Compact CIFAR-10 harder-data extension	20	decoder-input dim 20	Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport

Table S11: Effective dimensionality of the default neuron-wise broadcast in the main architectures. The implementation uses the output error vector only when layer width matches decoder output dimension; otherwise it falls back to a shared scalar. The main feedforward settings in this paper therefore test a genuinely rank-1 broadcast regime by default, which is why the higher-rank low-rank, path-transport, and pathway-vector controls are important.

Asset	Use in this paper	Access / license note
MNIST	Digit classification and derived nonnegative synthetic tasks	Public academic benchmark credited to the original dataset authors; used through standard public access, with no raw-data redistribution.
Fashion-MNIST	Apparel classification stress test	Public benchmark released with the Fashion-MNIST paper and repository; used through standard public access, with no raw-data redistribution.
CIFAR-10	Flattened harder-data diagnostic	Public research benchmark from the CIFAR dataset collection; used through standard public access, with no raw-data redistribution.
MICrONS	Biological reference statistics in the weight-distribution appendix	Public connectomics resource cited to the MICrONS Consortium; only aggregate comparison statistics are reported here.
Synthetic tasks	Context gating, noise resilience, cue integration	Generated procedurally from public benchmarks or random seeds described in Appendix E; no new third-party asset is introduced.

Table S12: Existing assets used in the manuscript. We credit the original sources in the bibliography, use public benchmark assets under their published access conditions, and do not redistribute third-party raw data in the anonymous submission.

Condition	MNIST		Fashion-MNIST	
	Exc σ	Inh σ	Exc σ	Inh σ
BP + Shunting	0.94 $\pm$ 0.01	0.87 $\pm$ 0.01	0.92 $\pm$ 0.01	0.84 $\pm$ 0.01
BP + Additive	1.30 $\pm$ 0.01	1.22 $\pm$ 0.01	1.16 $\pm$ 0.01	1.05 $\pm$ 0.01
Local + Shunting	1.22 $\pm$ 0.01	1.71 $\pm$ 0.02	1.13 $\pm$ 0.06	1.36 $\pm$ 0.06
Local + Additive	1.69 $\pm$ 0.16	1.57 $\pm$ 0.07	1.43 $\pm$ 0.16	1.11 $\pm$ 0.12
Song et al. 2005	$\sigma = 0.94$ (rat V1 L5, EPSP amplitude)			
MICrONS E $\rightarrow$ E	$\sigma = 1.14$ (mouse V1, synapse size, $n=3.9M$)			
MICrONS I $\rightarrow$ E	$\sigma = 0.92$ (mouse V1, synapse size, $n=3.5M$)			

Table S13: **Secondary weight-distribution summary.** Log-normal σ of excitatory and inhibitory synaptic weights (mean $\pm$ std across 5 seeds). Biological references are descriptive anchors only. With biologically constrained E/I ratios (4:1), shunting self-normalization produces weight heterogeneity in the range spanned by those references, whereas additive integration is broader and less stable.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state the paper’s main mechanistic claim, the scope of the conductance-based setting, and the regime-dependent nature of the empirical results; see Sec. 1 and Sec. 5.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The discussion explicitly narrows the scope of the claims and discusses limitations involving benchmark strength, broadcast assumptions, depth scaling, and the exploratory stress tests; see Sec. 5.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The voltage model, assumptions, theorem statements, and proof details are given in Sec. 2 and Appendix C; proof sketches and the exact factorization sanity check are cross-referenced in the main text.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The main paper and appendices specify the datasets, architectures, local-rule family, broadcast modes, representative hyperparameters, and compute regime needed to reproduce the headline results, and the supplementary ZIP provides anonymized code, configuration files, and reproduction instructions; see Sec. 4 and Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The anonymous submission includes an anonymized supplementary ZIP containing the model, training, diagnostic, and figure-generation code, representative configuration files, and reproduction instructions for the main experimental families. The public benchmarks are cited and accessed through their standard public sources; see Appendix B.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The main text states the benchmark families, seed protocol, representative local-rule settings, and major training choices, while the appendices and supplementary ZIP provide implementation details, configuration files, and operating-mode inventories; see Sec. 4 and Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The paper reports means and standard deviations across 5 seeds for the headline results and across 3 seeds for selected appendix-only support sweeps, states this explicitly in the text, and shows error bars or seed variability in the relevant tables and figures; see Sec. 4 and Appendix A.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix B describes the institutional GPU cluster setting, the use of single A100/H100-class GPUs, rough per-run runtime scales, and the fact that broader exploratory sweeps required additional compute beyond the final reported runs.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses standard public benchmarks and mechanistic simulations, does not involve human-subjects experimentation or restricted data collection, and we are not aware of any aspect that violates the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.

- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: The paper is a mechanistic study of dendritic credit assignment using public benchmarks and synthetic diagnostics, with no deployed system, human-subject data, scraped dataset, or direct societal application. The limitations section nevertheless warns against overgeneralizing the mechanism beyond the tested regimes; see Sec. 5.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The submission does not release a high-risk generative model, foundation model, or scraped dataset, so the safeguard question is not applicable here.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [\[Yes\]](#)

Justification: The paper credits all benchmark and connectomic assets, does not redistribute third-party raw assets, and includes an asset/access table with license or access terms where available; see Appendix B.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [\[Yes\]](#)

Justification: The only new released asset is an anonymized supplementary code ZIP. It includes reproduction instructions, representative configuration files, and the model/training/diagnostic scripts needed for the main experimental families; the paper does not release a new dataset, benchmark, or pretrained model.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [\[N/A\]](#)

Justification: The work does not involve crowdsourcing or direct research with human subjects.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not involve crowdsourcing nor research with human subjects.

- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The work does not involve crowdsourcing or human-subjects research requiring IRB review.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs are not part of the proposed method, theorem, experiments, or analysis pipeline, so no declaration is required for the core scientific contribution.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.